

May 6, 2026

11:01:07 AM

Starting parallel pool (parpool) using the 'Processes' profile ...
Connected to parallel pool with 12 workers.
Copying objective function to workers...
Done copying objective function to workers.

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Iteration	Iteration	Eval	Objective	Objective	BestSoFar	BestSoFar	SectionDepth	InitialLearn-
		workers	result	runtime	(observed)	(estim.)		Rate
=====								

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1	12	Best	0.802	137.56	0.802	0.802	3	0.71501
0.86017		0.0084489						
2	12	Best	0.762	158.61	0.762	0.76461	3	0.24648
0.96098		7.554e-07						
3	12	Accept	0.802	77.077	0.762	0.78867	3	0.29662
0.96327		1.8771e-06						
4	12	Accept	0.788	274.98	0.762	0.7885	3	0.47279
0.87449		2.6556e-05						
5	12	Accept	1	35.323	0.762	0.783	3	0.97776
0.80723		1.5617e-10						
6	12	Accept	0.788	45.481	0.762	0.78393	3	0.74587
0.90077		6.4405e-06						
7	12	Best	0.02	383.9	0.02	0.021432	1	0.071244
0.83028		3.408e-06						
8	12	Accept	0.05	424.6	0.02	0.026932	1	0.10281
0.87562		1.741e-05						
9	12	Accept	0.788	47.041	0.02	0.02012	1	0.81484
0.80347		3.4601e-09						
10	12	Accept	0.848	515.2	0.02	0.021013	2	0.22733
0.84905		0.00091996						
11	12	Accept	0.848	540.25	0.02	0.021082	2	0.53777
0.8041		1.0254e-06						

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12	12	Accept	0.15	473.95	0.02	0.019746	2	0.014337 ✓
0.88877	0.0048456							
13	12	Accept	0.848	647.99	0.02	0.019629	3	0.019656 ✓
0.81089	3.7567e-08							
14	12	Accept	0.17	658.29	0.02	0.020069	2	0.21196 ✓
0.94114	1.3204e-06							
15	12	Accept	0.42	665.96	0.02	0.020077	3	0.063416 ✓
0.95278	3.2691e-10							
16	12	Accept	0.148	701.92	0.02	0.020095	3	0.021625 ✓
0.93233	7.8154e-09							
17	12	Accept	0.292	288.95	0.02	0.020073	1	0.048534 ✓
0.92345	0.0077639							
18	12	Accept	0.384	781.74	0.02	0.02007	2	0.25803 ✓
0.94448	2.9357e-09							
19	12	Accept	0.078	420.71	0.02	0.020069	1	0.090615 ✓
0.97982	1.4075e-10							
20	12	Accept	0.762	584.63	0.02	0.025587	3	0.15617 ✓
0.95338	1.2172e-07							
=====								✓
=====								
Iter	Active	Eval	Objective	Objective	BestSoFar	BestSoFar	SectionDepth	InitialLearn-
Momentum	L2Regulariza-							
	workers	result		runtime	(observed)	(estim.)		Rate
tion								
=====								✓
=====								
21	12	Accept	0.024	320.5	0.02	0.025763	1	0.024036 ✓
0.92806	6.3606e-07							
22	12	Accept	0.8	13.696	0.02	0.024679	1	0.99069 ✓
0.97869	3.3851e-07							
23	12	Accept	0.372	296.91	0.02	0.024884	1	0.010453 ✓
0.97894	0.008037							
24	12	Accept	0.848	592.98	0.02	0.024853	3	0.17046 ✓
0.81337	0.00036647							

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25	12	Accept	0.054	412.65	0.02	0.023366	1	0.057384 ✓
0.9709	3.2962e-05							
26	12	Accept	0.02	350.35	0.02	0.020254	1	0.057495 ✓
0.87918	1.0095e-10							
27	12	Accept	0.084	312.84	0.02	0.020507	1	0.010306 ✓
0.80954	0.0011199							
28	12	Accept	0.022	300.17	0.02	0.017793	1	0.027702 ✓
0.84685	2.5735e-06							
29	12	Accept	0.02	310.97	0.02	0.0083454	1	0.049192 ✓
0.89069	8.3214e-08							
30	12	Accept	0.074	390.69	0.02	0.0052231	1	0.13288 ✓
0.92952	3.6225e-09							

Optimization completed.

MaxObjectiveEvaluations of 30 reached.

Total function evaluations: 30

Total elapsed time: 1223.1157 seconds

Total objective function evaluation time: 11165.9112

Best observed feasible point:

SectionDepth	InitialLearnRate	Momentum	L2Regularization
1	0.071244	0.83028	3.408e-06

Observed objective function value = 0.02

Estimated objective function value = 0.025738

Function evaluation time = 383.9008

Best estimated feasible point (according to models):

SectionDepth	InitialLearnRate	Momentum	L2Regularization

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```
1          0.049192      0.89069      8.3214e-08
```

```
Estimated objective function value = 0.0052231
Estimated function evaluation time = 271.7651
```

```
acc_25dB =
```

```
0.9700
```

```
acc_20dB =
```

```
0.7176
```

```
acc_15dB =
```

```
0.4008
```

```
acc_10dB =
```

```
0.2654
```

```
>>
```